

YASHASWINI VISARAPU

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PROFESSIONAL SUMMARY

Integrated B.Tech + M.Tech student in Electronics and Communication Engineering specializing in Communication and Signal Processing. Focused on embedded systems and firmware development using Embedded C. Experienced in building interrupt-driven systems, UART communication modules, cooperative task schedulers, state machines, and machine health monitoring prototypes.

EDUCATION

Jawaharlal Nehru Technological University, Hyderabad (JNTUH)

Integrated B.Tech + M.Tech in Electronics and Communication Engineering (Communication and Signal Processing) | 2023–2028

TECHNICAL SKILLS

Programming: C, Embedded C, Python

Embedded Concepts: UART, Interrupts (ISR), Ring Buffers, State Machines, Cooperative Scheduling

Tools: PlatformIO, Wokwi, Git, VS Code

Hardware: NodeMCU, Arduino Uno

Academic: Digital Signal Processing, Communication Systems, Digital Electronics

PROJECTS

UART Command Shell

Developed an interrupt-driven UART command shell in Embedded C for runtime interaction and debugging. Implemented command parsing and fixed-size buffering for asynchronous serial communication. GitHub: github.com/Yashaswini-200/uart-command-shell

Event-Driven UART System

Designed an event-driven communication framework using UART interrupts and producer-consumer principles. Separated ISR and main-loop responsibilities to reduce polling overhead. GitHub: github.com/Yashaswini-200/Event-driven-UART-command-system

Cooperative Task Scheduler

Built a timer-based cooperative scheduler using function pointers for periodic task execution without an RTOS. Organized tasks using a fixed task table and non-blocking execution flow. GitHub: github.com/Yashaswini-200/cooperative-scheduler

Predictive Maintenance System

Developed a machine health monitoring prototype using vibration RMS values, temperature monitoring, and a fault-detection state machine. Simulated and tested the system using NodeMCU, PlatformIO, and Wokwi. GitHub: github.com/Yashaswini-200/Predictive-Maintenance-System

FIR Filter Architecture Comparison

Compared basic, symmetric, pipelined, and pipelined symmetric FIR filter architectures and analyzed trade-offs in implementation complexity and throughput. GitHub: github.com/Yashaswini-200/fir-filter-architecture-comparison

INTERESTS

Embedded Systems, Firmware Development, Signal Processing, Photography, Video Editing